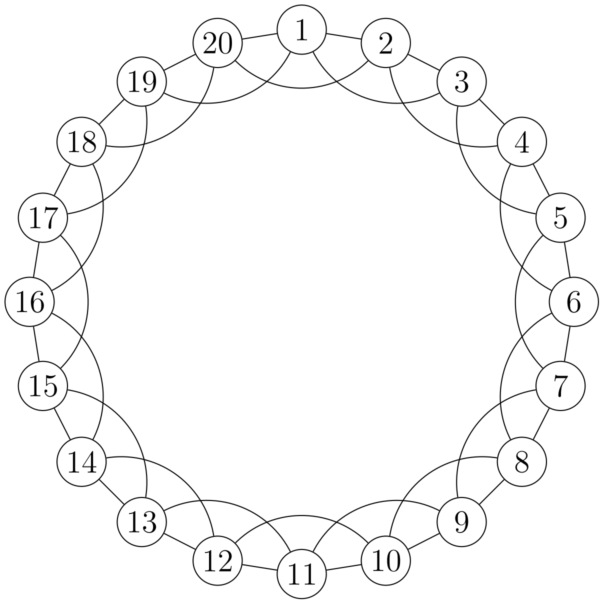
Turning the Dial

Ringville has grown. Twenty houses now sit around the ring, and people have become slightly more sociable: everyone knows the *two* neighbors on each side. Nobody else.



You now have names for everything you measured before: distance, average path length, diameter, clustering, the random baseline, the small-world index σ. This sheet is where you find out how much those names are worth.

Part 1: The town before anything is rewired

Question 1: Measure the town as it stands.

1. Write down the four people that person 1 knows: ____________________
2. How many pairs among those four are themselves connected? ________

   So person 1's local clustering coefficient is ________
3. Every person in this town sits at the corner of the same number of triangles. How many triangles does the whole town contain? ________

   Say in one line how you got there without drawing all of them.

   ____

Question 2: Now distances. Instead of listing all nineteen people, use the ring: a person d houses away along the circle needs how many handshakes? Fill the top row by walking a few examples on the drawing, then finish the row by the pattern.

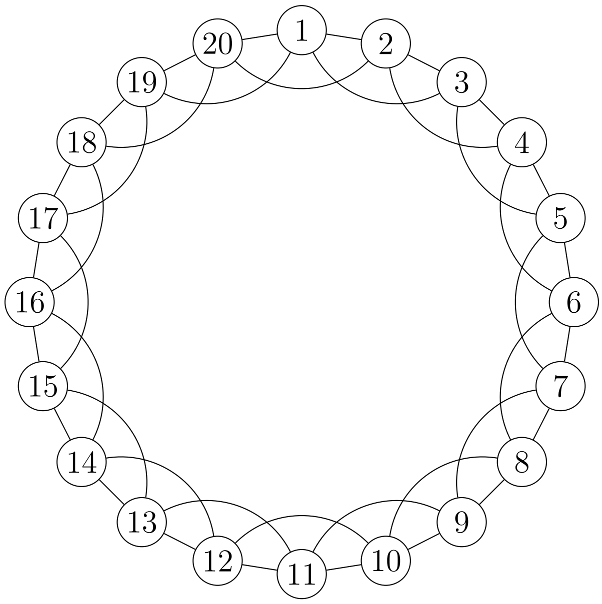
Houses away d	1	2	3	4	5	6	7	8	9	10
Handshakes										
How many people										

1. Average handshakes from person 1 to everyone else: ____________
2. Diameter of the town: ________
3. Does the average you just computed depend on *which* person you started from? Why?

   ____

Part 2: One edge

Question 3: Person 1 falls out with person 2 and befriends person 11 instead. So: **erase the edge 1–2**, and **draw the edge 1–11**. The town still has exactly forty friendships — you moved one, you did not add one.



Work outward from person 1 in layers: first everyone one handshake away, then everyone reachable from those, and so on. Fill in the counts.

Handshakes	1	2	3	4	5
How many people					

The counts must add up to ______. Do they?

Question 4: Fill in the ledger for that single moved friendship.

	Before	After	Change (%)
Friendships changed	—	1 out of 40	
Avg. handshakes from person 1			
Farthest person from 1			
Triangles in town			

For the triangle row, you do not need to recount the town. Ask instead: which triangles used the edge 1–2, and does the new edge 1–11 create any?

1. Triangles killed by removing 1–2: ______ Triangles created by adding 1–11: ______

2. You changed 2.5% of the friendships. Which of the two quantities — distance or triangles — moved by much more than 2.5%?

   ____

3. Why is the new edge so much better at shortening paths than the old edge 1–2 was? Answer in terms of what each edge lets you skip.

   ____

Part 3: Turning the dial

Now do it the way the model does. Go through all forty friendships one at a time, and with probability p move that friendship to a randomly chosen person. $p = 0$ leaves the town untouched; $p = 1$ scrambles it completely.

Question 5: Pick any one triangle in the original town.

1. How many friendships does it consist of? ______
2. That triangle is still there after the rewiring only if

   ____

3. So the chance it survives is ____________, and therefore the town's clustering after rewiring is

$$C(p) \;=\; C(0) \times \underline{\hspace{3cm}}$$

4. Evaluate:

   $C(0.01) =$ ________ $C(0.1) =$ ________ $C(0.5) =$ ________

Question 6: Now the other quantity. At $p = 0.01$, how many of the forty friendships get moved, on average? ________

From Question 4 you know what one moved friendship does. So:

1. Roughly how many people does *one* shortcut bring closer? ____________
2. Roughly how many triangles does *one* moved friendship destroy?

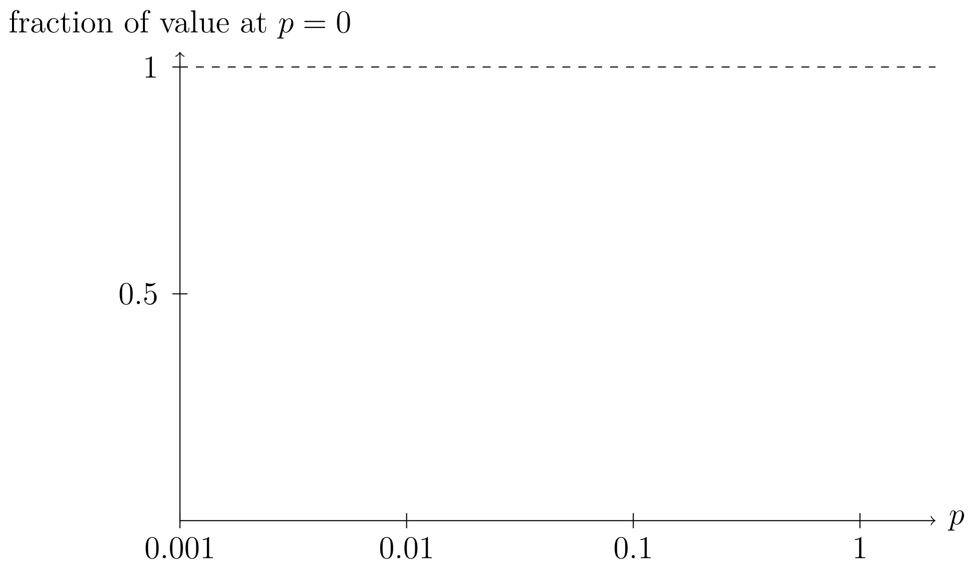
   ____

3. One of these two effects is felt by the whole town, the other only near the edge that moved. Which is which, and why does that make the two curves fall at completely different values of p?

   ____

   ____

Question 7: Sketch both curves on the same axes: clustering as a fraction of its starting value, and average handshakes as a fraction of its starting value. The horizontal axis multiplies by ten at each tick.



Shade the band of p where the town has short chains *and* still keeps most of its triangles. Then answer: is that band narrow, so that a town would have to be tuned carefully to land in it, or wide enough that almost any real town falls in by accident?

Part 4: The number that lies

You built σ yourself on the previous sheet: how much more clustered the town is than chance, divided by how much longer its chains are than chance.

For a coin-flip town of n people with $\langle k \rangle$ friends each: clustering $\approx \langle k \rangle/(n-1)$ and average handshakes $\approx \ln n / \ln\langle k \rangle$. Useful values: $\ln 4 \approx 1.4$, $\ln 20 \approx 3.0$, $\ln 10{,}000 \approx 9.2$.

Question 8: Compute σ for Ringville *before* any rewiring — the plain ring you measured in Part 1, with $p = 0$.

	Ringville ($p = 0$)	Coin-flip town	Ratio
Clustering			
Avg. handshakes			

$\sigma =$ ____________

Is it above 1? ______ But this town has no shortcuts at all. What has just gone wrong?

____

Question 9: Maybe the trouble is that twenty is a small town. Test that. Keep the same ring rule — everyone knows two neighbors on each side — but let the town have 10,000 people. You do not need exact numbers; a rough value for each cell is enough.

	$n = 20$	$n = 10{,}000$	Grows or shrinks?
Clustering of the ring			
Clustering of coin-flip town			
Avg. handshakes, ring			
Avg. handshakes, coin-flip			
σ			

(For the ring, a person's average distance is about $n/8$ — convince yourself from the table in Question 2 before you use it.)

1. As the ring town grows, does σ approach 1, stay put, or run away? ____________

2. So what does "$\sigma > 1$" actually certify?

Question 10: A paper lands on your desk. It reports $\sigma = 2.3$ for a brain network and concludes it is a small world.

1. Name two things the paper must tell you before that number means anything.

2. σ compares the network to a coin-flip town only. From Questions 8 and 9 you know why that is not enough. Propose a second comparison that would have caught the problem, and say what value your new test should give for a plain ring.